

Upgrading Santander's Active Directory domain controllers

How an in-situ server upgrade and a functional level promotion gave a financial-services directory a secure, stable footing for future scale and compliance.

Period	My role	Platform	Upgrade to
March to December 2017	Project lead	Active Directory domain controllers	Windows Server 2008 R2

Overview

Between March 2017 and December 2017, I led a critical infrastructure upgrade of Santander's Active Directory Domain Controller (AD DC) environment.

The goal was to modernise the platform so it could support future scalability, security and compliance requirements across the organisation's financial-services operations.

The approach

The work had two stages. First, the existing domain controller servers were upgraded in place to Windows Server 2008 R2. Then the AD DS forest and domain functional levels were promoted, which aligned the directory with Microsoft's enterprise standards and enabled more advanced identity-management capabilities.

What was delivered

Server OS upgrade	Moved legacy domain controller servers to Windows Server 2008 R2, improving performance and patch compliance.
Functional level promotion	Raised the AD DS forest and domain functional levels to unlock enhanced authentication, replication and policy features.
Risk mitigation	Put rollback and validation plans in place to ensure zero downtime during upgrade windows.
Testing and validation	Ran pre-production testing, schema verification and replication health checks across multiple sites.
Stakeholder coordination	Worked with infrastructure, security and compliance teams to align technical changes with Santander's governance framework.

Outcome

The upgrade delivered a secure, stable and future-ready Active Directory environment. It improved operational resilience and made it possible to integrate with newer Microsoft technologies later.

It also strengthened Santander's ability to manage identity and access across its global financial network.

Impact

- Enhanced security posture and compliance alignment.
- Improved directory performance and replication efficiency.
- A foundation for future cloud and hybrid identity initiatives.